

Problem 2 (13 marks)

A country produces **agricultural products (A)** and **electronics (E)** with the production possibility frontier (PPF):

$$A^2 + 2E^2 = 900.$$

The world relative price is $P_E/P_A = 3$. The country consumes at the point where preferences satisfy $E = 0.5A$.

- (1) Determine the production point (A_P, E_P) that maximizes the value of output at world prices.
- (2) Find the consumption bundle (A_C, E_C) , identify which good is exported, and compute the value of exports and imports.
- (3) Verify that trade is balanced at world prices.
- (4) Explain in words how an increase in P_E/P_A (improvement in the terms of trade) affects the country's production, consumption, and welfare.
- (5) Suppose the economy experiences **export-biased growth** that expands production possibilities more in electronics than in agriculture.
 - Describe how this changes the country's trade pattern and the direction of the terms-of-trade effect.
 - Discuss whether welfare necessarily increases after such growth.